

We see the system is consisted of three main parts the control subsystem which includes the feedforward lookup tables (the dominant force) method in addition to the auxiliary PI controller the combination of these two is called the hybrid

feedforward controller, the Joule heating system that is provided with temperature signal from the hybrid controller to reach the desired curvature or track the desired trajectory, and the plant SMA bending wire that is bent due to load and contract and submit to the load due to the temperature it exhibits, I shall mention that I treat the previously extracted 1-D lookup tables that was provided by FORC measurement method was at a load that is bending the tip of the wire to 52 m^{-1} as the baseline of our experiment and any deviation above or under it will be treated with the auxiliary PI controller, also the auxiliary PI controller has upper limit with 30 K signal and lower limit with -30 K in addition to anti windup clamping method.

3. Load varying cases

From now on I will make my experiment testing the system limits, first by experimenting the system at the recognized 52 m^{-1} load then, I will increase and decrease the load.

Case (a) the load at 52 m^{-1} load baseline

In this case I test the hybrid feedforward controller and see what changes it makes to the system with including PI auxiliary controller to wipe out the remaining error

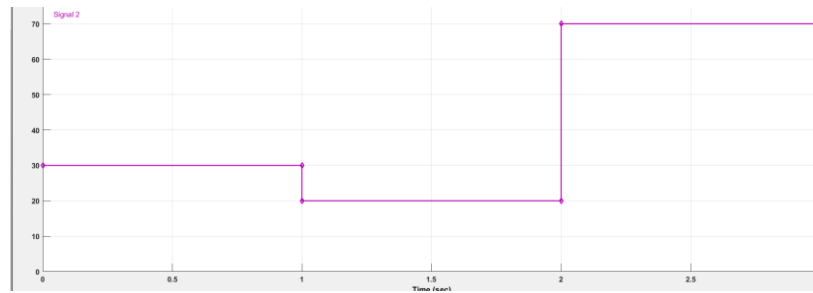


Figure (2) The input desired trajectory

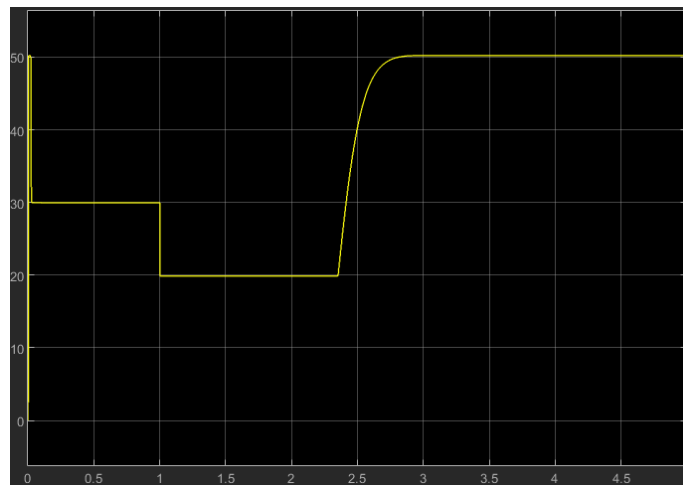


Figure (3) The output curvature

We first see the wire was bent to 52 m^{-1} due to the load, the 30 m^{-1} step took a place, afterward the 20 m^{-1} step happened and the system couldn't exceed his physical limit to reach 70 m^{-1} step.

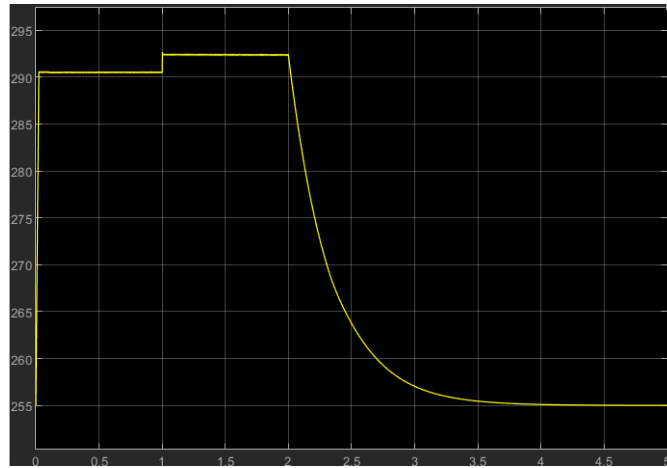


Figure (4) the temperature curve

In figure (4) we see here there is a spike at 20 m^{-1} step proving that the auxiliary PI controller was trying to be the dominant driving force to reach the desired destination, afterward it had a cut down after the destination was achieved, then again it fought with its full power to achieve that 70 m^{-1} step but got hit by the physical limit of the system at 52 m^{-1} , you should notice the cut down didn't change anything because the wire was not going to change phases with this tiny drop of the temperature.

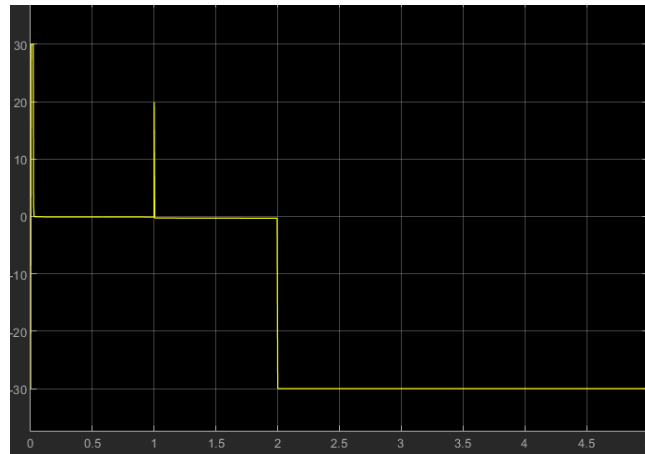


Figure (5) The negative gain auxiliary PID controller output

We see here where the spikes came from, the PI controller was providing signals to Joule heating system with its full power and as soon as the desired curvature was achieved afterward the cut down occurred then when the 70 m^{-1} step signal happened it jumped to its maximum power again but was continuing to operate at it to end of simulation because the wire hit its physical limit and cannot be bent more so the error was still there (the load only bent the wire to 52 m^{-1} you can't simply go beyond that).

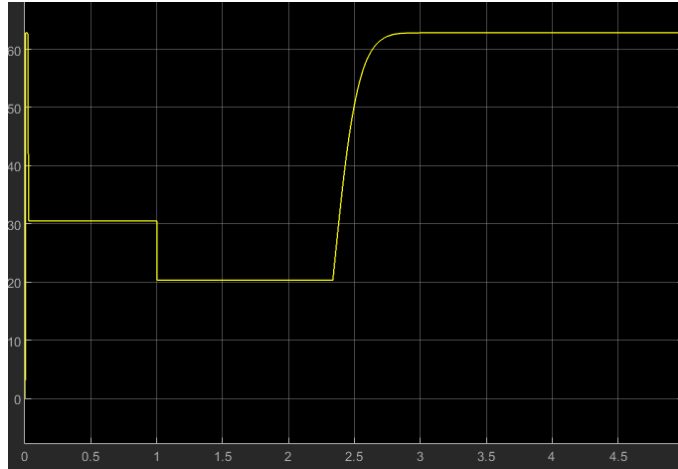


Figure (6) the output curvature at increased load experiment

Case (b) Increasing the load above 52 m^{-1} load baseline

We see the wire was bent more above 60 m^{-1} then how it went along the desired trajectory well until it hit the physical limit of the wire at more than 60 m^{-1} .

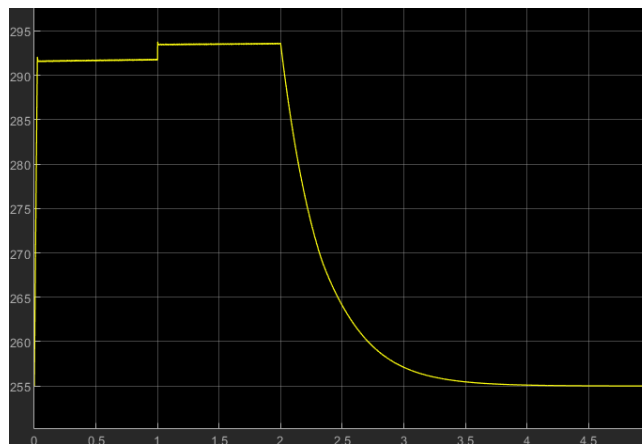


Figure (7) The temperature curve

We see a further increase in the targeted temperature due to the load increasing by 1-2 K (as a proof that the model considers the stress-dependent temperatures) and the PI controller spikes appeared in each of the heating steps.

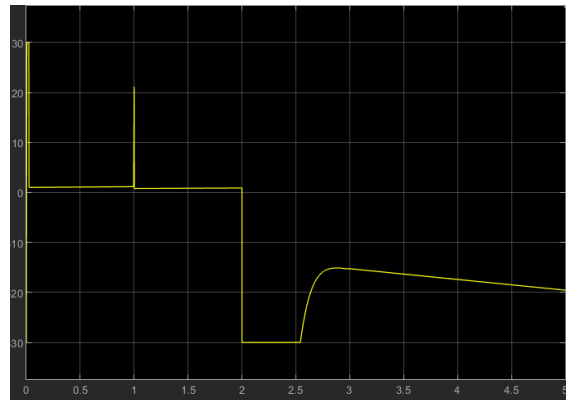


Figure (8) The negative Gain PI controller output

Here is the contributions of the auxiliary PI controller, it was confused due to the physical limit (windup), so I tuned PI gains.

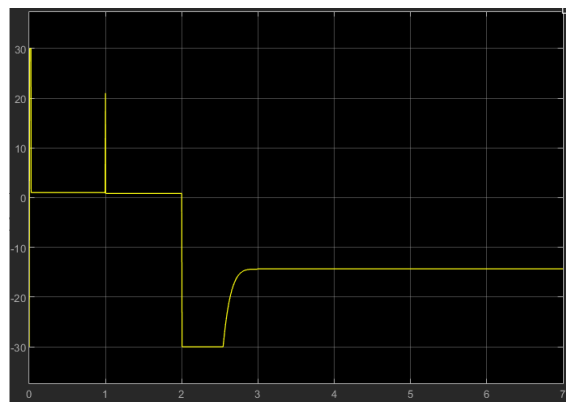


Figure (9) The negative Gain PI controller after Tuning

After tuning the integral part of the PI controller it settled around -15 K, it was still fighting to exceed the physical limit but it hit both the thermal limit at 255 K and physical limit, at first we see here how the PI controller tried to be the dominant part of the controller but always got that saturation bounds blocking its way.

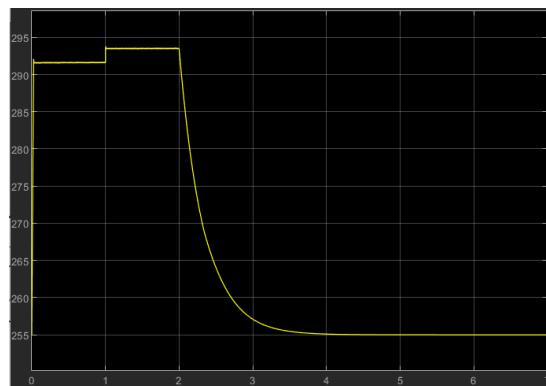


Figure (10) the temperature curve after tuning

It didn't change in the course of the 7 seconds simulation, I needed to make sure that there is no steady state error exist at the end due to PI controller and as long as it settled at -15K and didn't change with the time. Here we tested increasing the load and saw how the hybrid controller dealt with it and tested the limits of this case.

Case (c) decreasing the load under the 52 m^{-1} load baseline

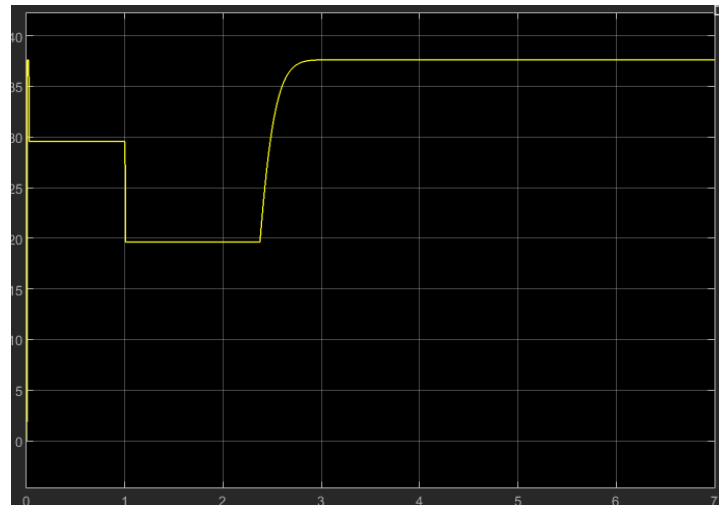


Figure (11) the output curvature with decreased load

Here we see the load bent the wire to 37 m^{-1} approximately, the 30 m^{-1} step happened then the 20 m^{-1} step happened then it hit the physical limit in its way to 70 m^{-1} , also there is a small tracking error around 1 m^{-1} at 20 m^{-1} step and 30 m^{-1} step

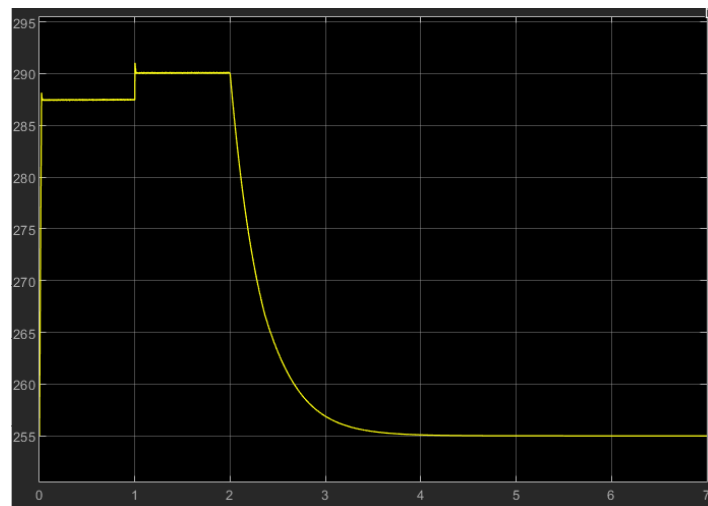


Figure (12) the temperature curve

Here also appeared the stress –dependent temperature as the load decreased the amount of heat to reach certain destination is decreased thanks to the auxiliary PI controller.

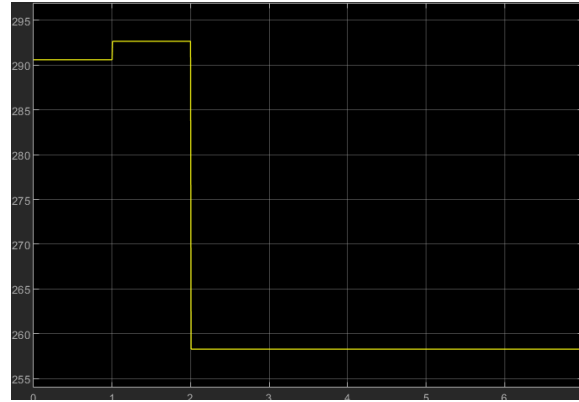


Figure (13) The lookup tables output

By comparing the two plots [Figure (12) and Figure (13)] we notice the effect of the auxiliary PI controller modifying the input signal

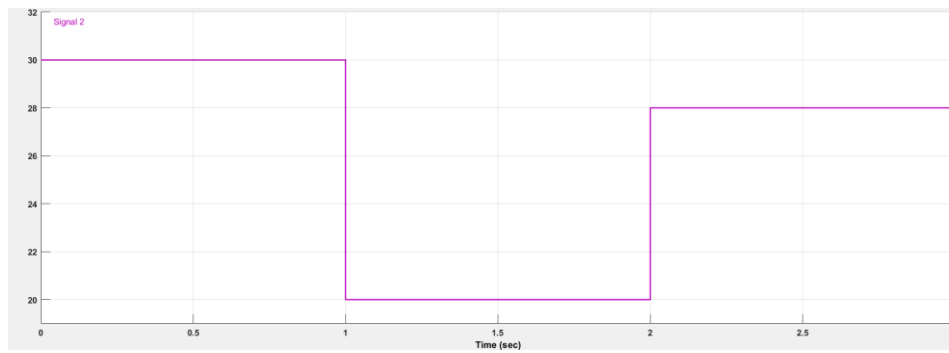


Figure (14) The inbounds trajectory in decreased load case

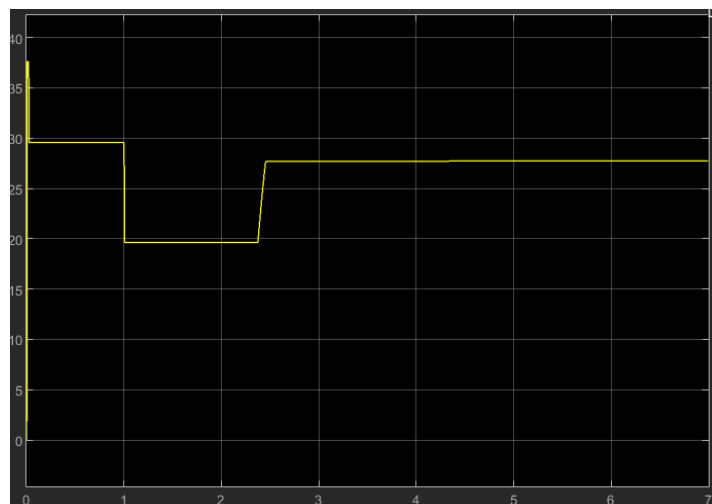


Figure (15) the output curvature of the inbounds desired trajectory

The tracking here is doing well with tiny consistent error less than 1 m^{-1} we can tune the PI controller more to achieve more efficient tracking but as long as the error being this small it's accepted.

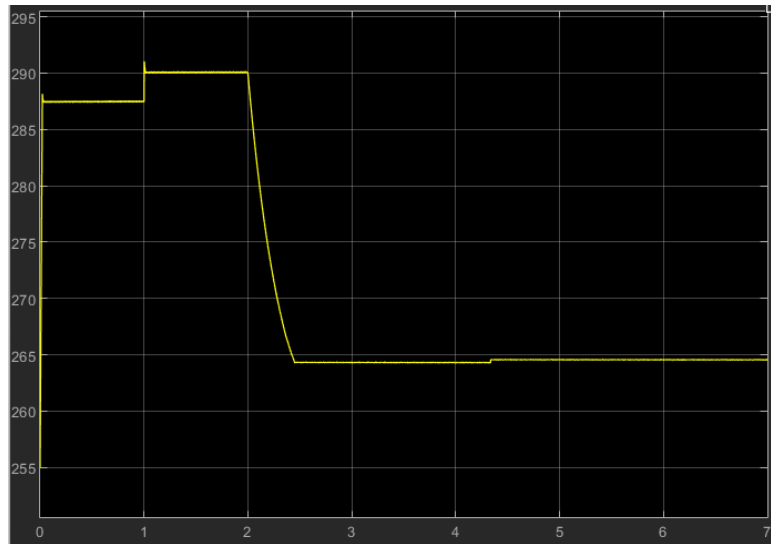


Figure (16) The temperature curve

We see the small increase in the temperature at the end but it doesn't affect the system as long as there no further increase in it making it to change phases (here to go from martensite to austenite phase)

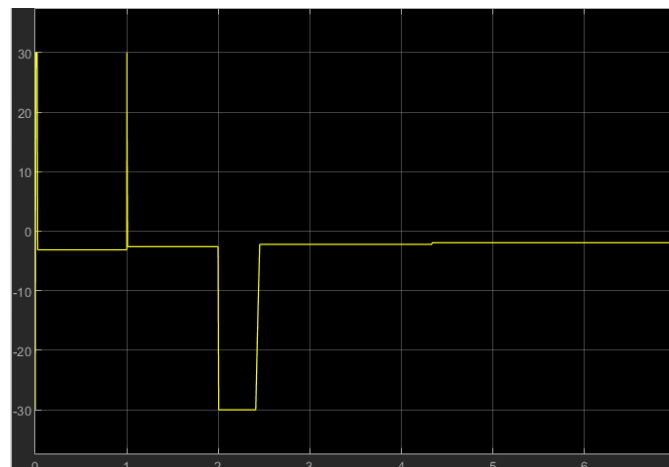


Figure (17) The negative gain PI controller output

Here we see where the small increase in the temperature came from but as long as it doesn't affect the system it doesn't matter.

4. Safety operating and the balancing between the electric power and the cooling method

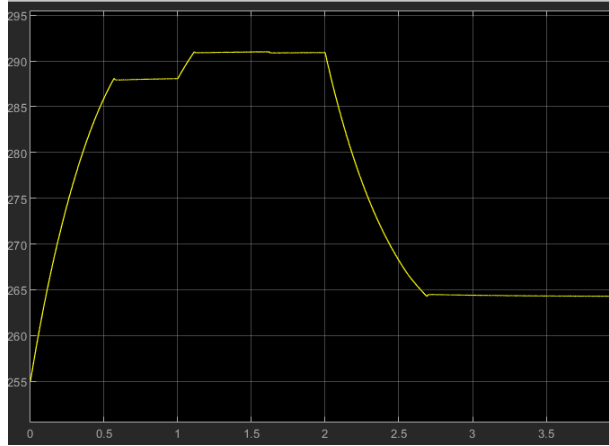


Figure (18) The temperature output with safety consider at 3 V operating

We see here how the temperature is raising much slower in the first step lagging .5 second and also doesn't jump instantly to the next step, we said in the previous documentation that after the application is on it should go close to austenite starting phase even without commands given to terminate the first step delay.

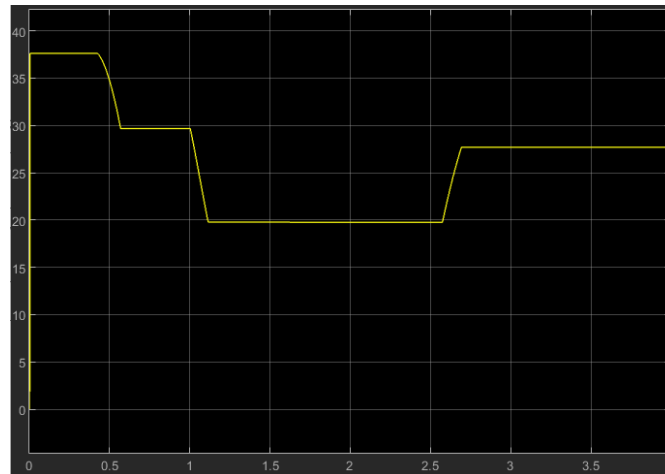


Figure (19) the output curvature

We see here the smooth tracking and the stability provided with the auxiliary PI controller without dealing with inner loop temperature controller, and that is for later appearing problem that happened when I decrease the input applied voltage to safety requirements in the previous documentation also taking in consideration we would change the cooling method in real life, it appears that hybrid feedforward controller will deal with it well.

5. The conclusion

During the course of this series I wanted to discover an applicable and practical way to control the SMA wire and predict the problems that would appear in front of me dealing with the real application, I have shown how the convection would be the main issue in the real environment with the delay it plays a big part governing it during the cooling step in specific also we should keep the cooling gain(convection method) balanced with the input electric power that we should safely use in my situation, I tested how the system operates first with high electric power that can harm the wire in experiment to treat the cooling method issue first then we balance the system to get the best results without damaging the wire, afterwards we tested the hybrid feedforward controller by making baseline FORC extracted two 1-D lookup tables and making them doing the heavy work then comes the role of the auxiliary PI controller of handling the small deviation also dealing with the tracking error, I hope this work will be the base of successful real-life experiment by relating the model based design to the real life practical not just to be a theoretical

work for that I considered the real environment physics limitations in the work and the hardships in data extraction also were discussed in the topic, this series was discussing this nonlinear topic in simple way and dealt with every problem as a standalone case and testing everything individually from the plant itself through the joule heating and convection, lastly dealing with the changing of the bending load and how the auxiliary PI controller compensate the different cases of it .